AI ASSISTED CODING ASSIGNMENT – 9.5

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BATCH-02

Problem 1: String Utilities Function

Consider the following Python function:

def reverse_string(text):

return text[::-1]

Task:

1. 1. Write documentation in:

   (a) Docstring

   (b) Inline comments

   (c) Google-style documentation

2. Compare the three documentation styles.
3. Recommend the most suitable style for a utility-based string library.

Programs > Python > AIAC > demo.py > ...

C:\Users\DELL\OneDrive\Desktop\Programs

```python
    """
    Docstring for the reverse_string function.
    :param text: The string to be reversed.
    :return: The reversed string.
    """
    return text[::-1]
if __name__ == "__main__":
    input_string = "Hello, World!"
    reversed_string = reverse_string(input_string)
    print(f"Original string: {input_string}")
    print(f"Reversed string: {reversed_string}")
```

```python
def reverse_string(text):
    return text[::-1] # This line reverses the input string by slicing it with a step of -1.
if __name__ == "__main__":
    input_string = "Hello, World!" # This is the string that we want to reverse.
    reversed_string = reverse_string(input_string) # We call the reverse_string function and store the
    # result in the variable reversed_string.
    print(f"Original string: {input_string}") # This line prints the original string.
    print(f"Reversed string: {reversed_string}") # This line prints the reversed string.
```

```python
def reverse_string(text: str) -> str:
    """
    Docstring for the reverse_string function.

    :param text: The string to be reversed.
    :type text: str
    :return: The reversed string.
    :rtype: str
    :exceptions: None
    :error handling: None
    :side effects: None
    :raises: None
    """
    return text[::-1]
if __name__ == "__main__":
    input_string = "Hello, World!"
    reversed_string = reverse_string(input_string)
    print(f"Original string: {input_string}")
    print(f"Reversed string: {reversed_string}")
```

Output:

```
PS C:\Users\DELL\OneDrive\Desktop\Programs> python -u "c:\Users\DELL\OneDrive\Desktop\Programs\Python\
Original string: Hello, World!
Reversed string: !dlroW ,olleH
Original string: Hello, World!
Reversed string: !dlroW ,olleH
PS C:\Users\DELL\OneDrive\Desktop\Programs>
```

Problem 2: Password Strength Checker

Consider the function:

def check_strength(password):

return len(password) >= 8

Task:

1. Document the function using docstring, inline comments, and Google style.
2. Compare documentation styles for security-related code.
3. Recommend the most appropriate style.

Output:

Problem 3: Math Utilities Module

Instructions:

1. 1. Create a module math_utils.py with functions:
   square(n)
   cube(n)
   factorial(n)
2. Generate docstrings automatically using AI tools.
3. Export documentation as an HTML file.

Output:

Problem 4: Attendance Management Module

Task:

1. Create a module attendance.py with functions:
   mark_present(student)
   mark_absent(student)
   get_attendance(student)
2. Add proper docstrings.
3. Generate and view documentation in terminal and browse

Output:

Problem 5: File Handling Function

Consider the function:

```
def read_file(filename):

with open(filename, 'r') as f:

return f.read()
```

Task:

1. Write documentation using all three formats.
2. Identify which style best explains exception handling.
3. Justify your recommendation.